Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport
la
Lucrarea de laborator Nr. 4

Disciplina: Tehnici de programare

Tema: ” Introducere în limbajul Python”

A efectuat : st. gr.SI – 261 Talchiv Corina

A verificat : asistent universitar Dumitrașcu Marius

Chișinău – 2026

SARCINA LUCRĂRII

În acest laborator este nevoie de parcurs din două site-uri Codedex și Learnpython.org și de trecută fiecare sarcină împarte. Pe learnpython.org trebuie să accesezi fiecare capitol din meniu (precum *Dictionaries*, *Generators*, *Regular Expressions* sau *Partial functions*) și să completezi exercițiile de cod din editorul interactiv apăsând pe butonul Submit. Pe Codédex trebuie să creezi un cont gratuit, să intri în cursul de Python și să rezolvi provocările practice din fiecare lecție, trecând prin pașii teoretici și scriind codul cerut în consolă până la finalizarea tuturor modulelor.

Modulul : Hello World

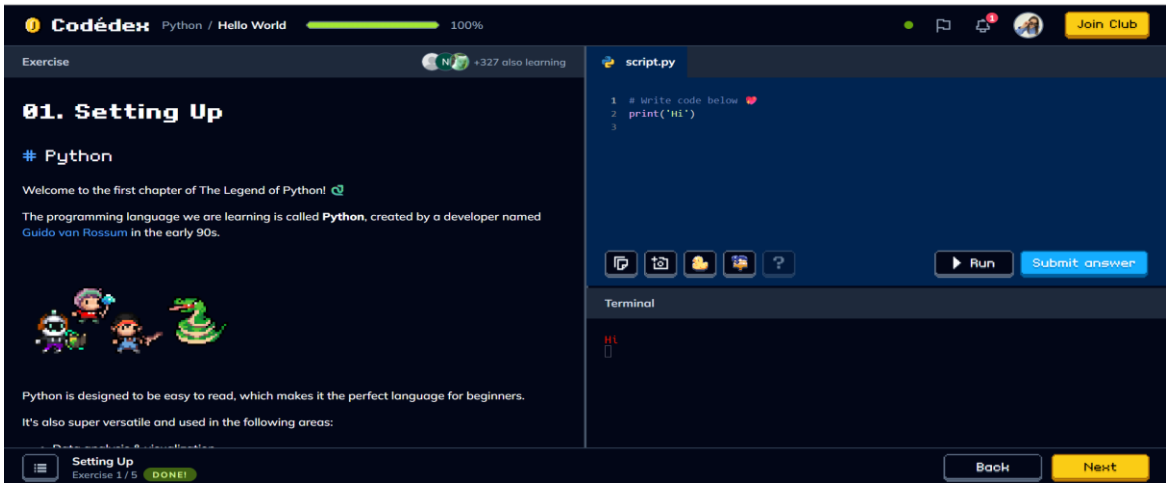


Figura 1 – Prima line de cod în python

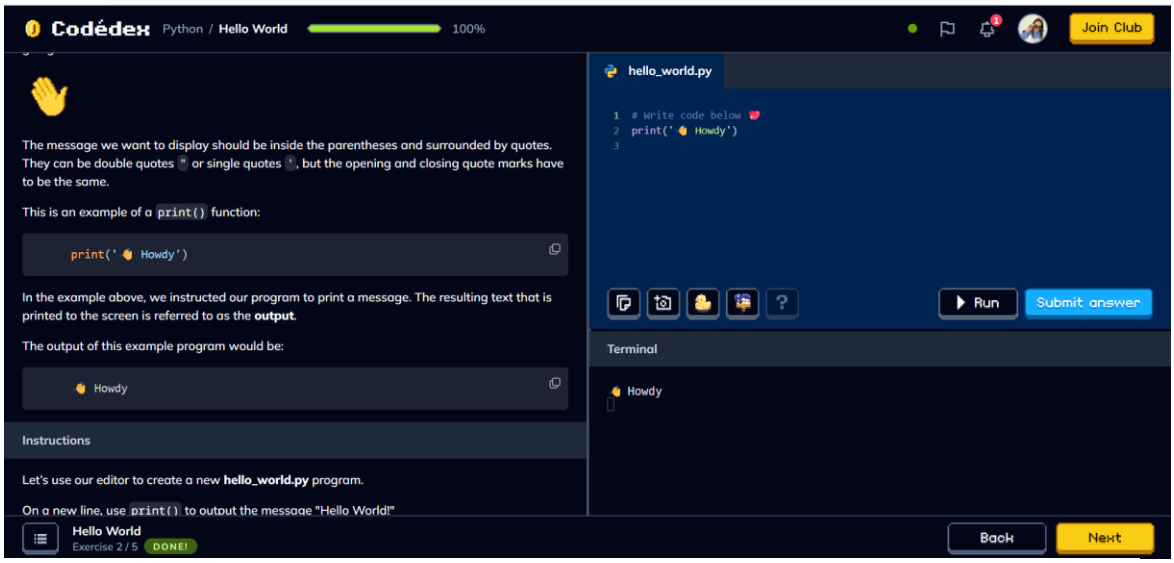


Figura 2 – Mesaj de salutare



Figura 3 – Un tringhi realizat



Figura 4 – Realizarea inițialilor noastre personale

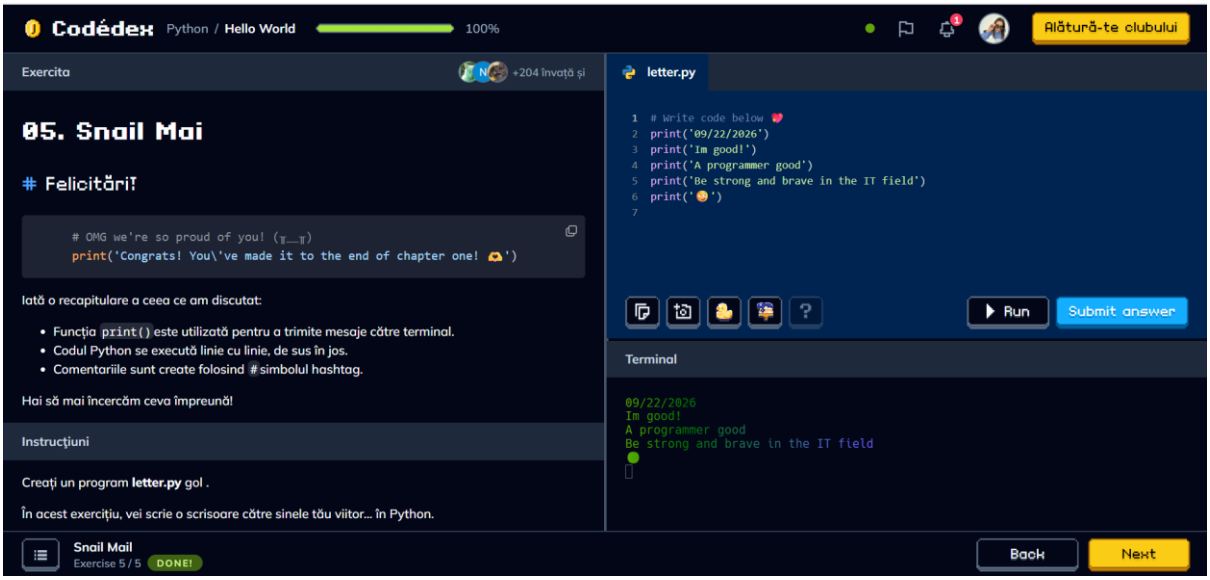


Figura 5 – Descriere personală

Modul Variabile:

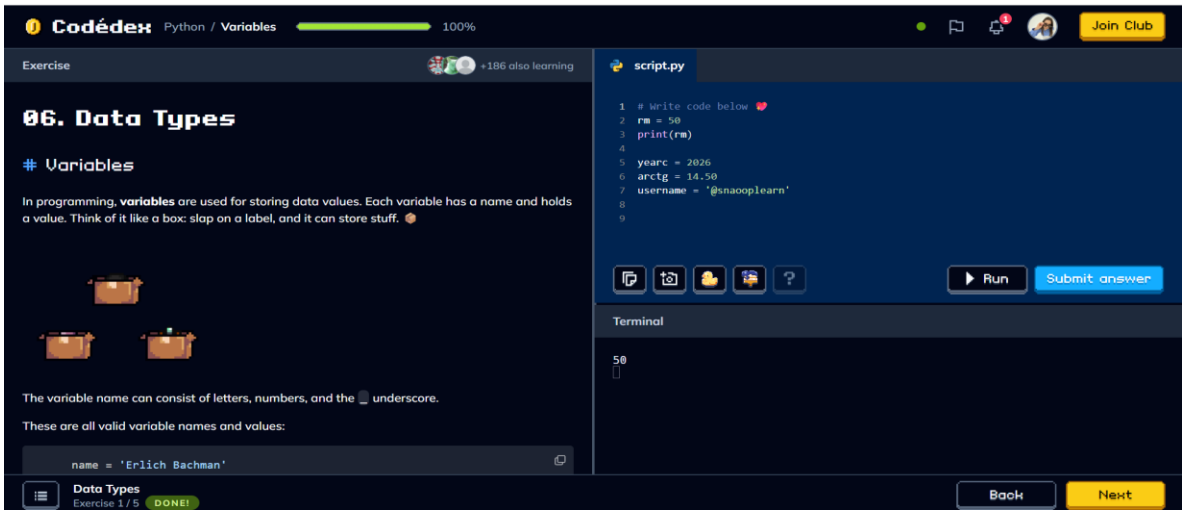


Figura 6 – 4 variabile de utilizat

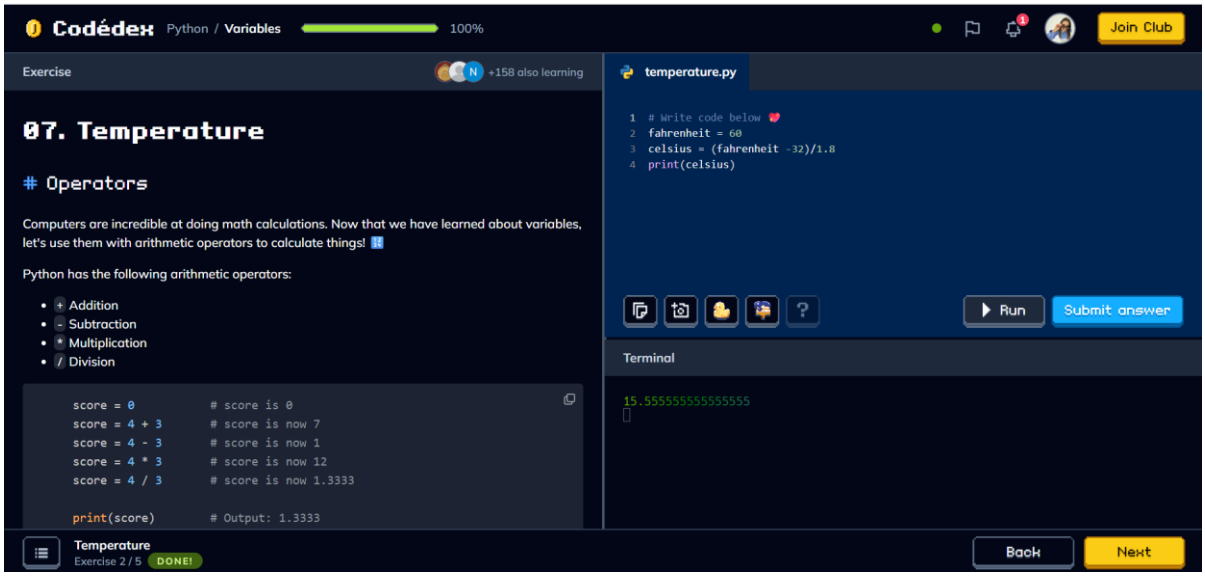


Figura 7 – Calcularea temperaturii

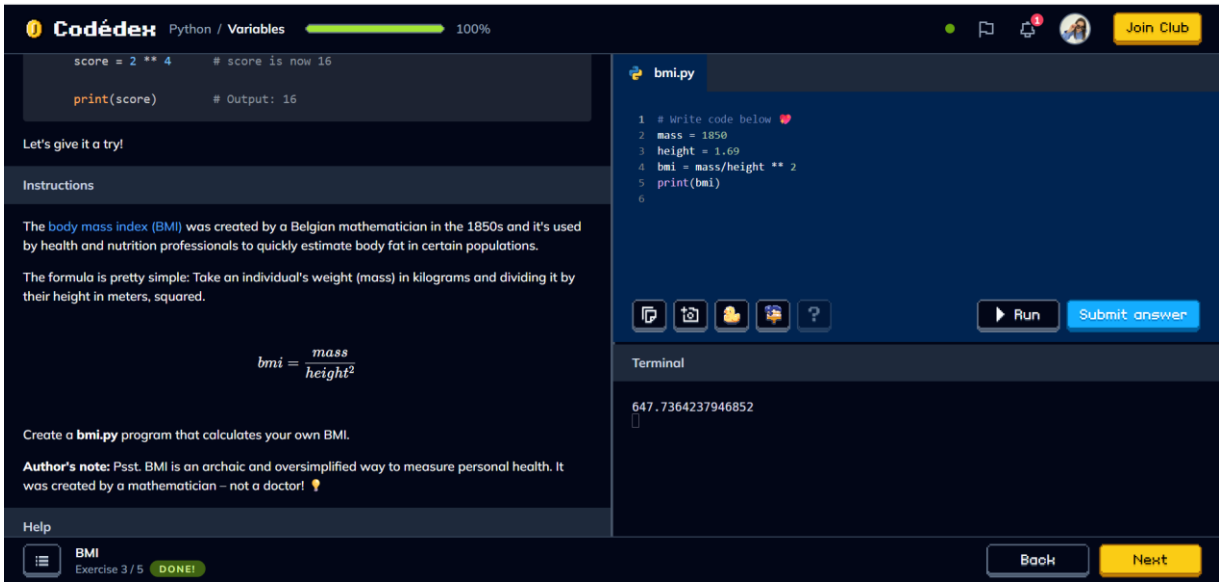


Figura 9 – Determinarea masa musculară

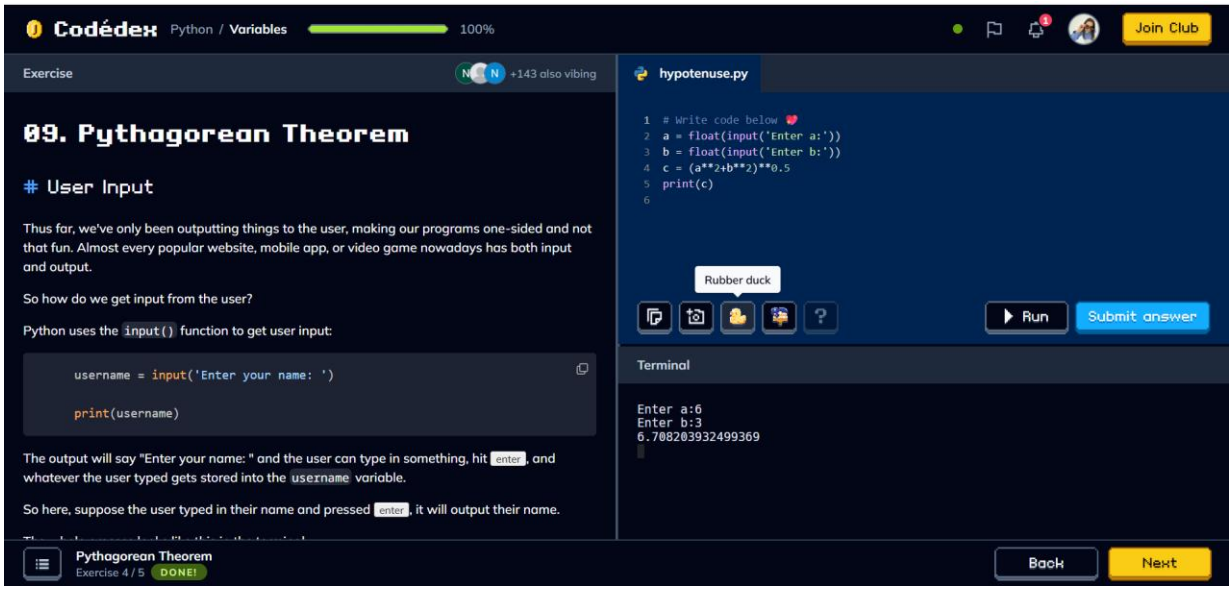
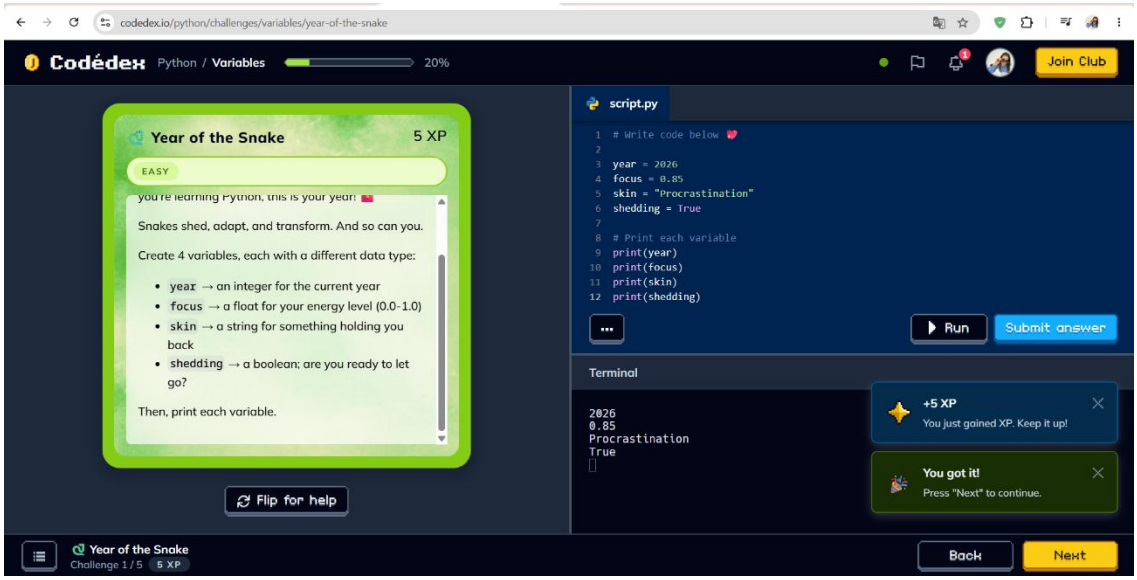


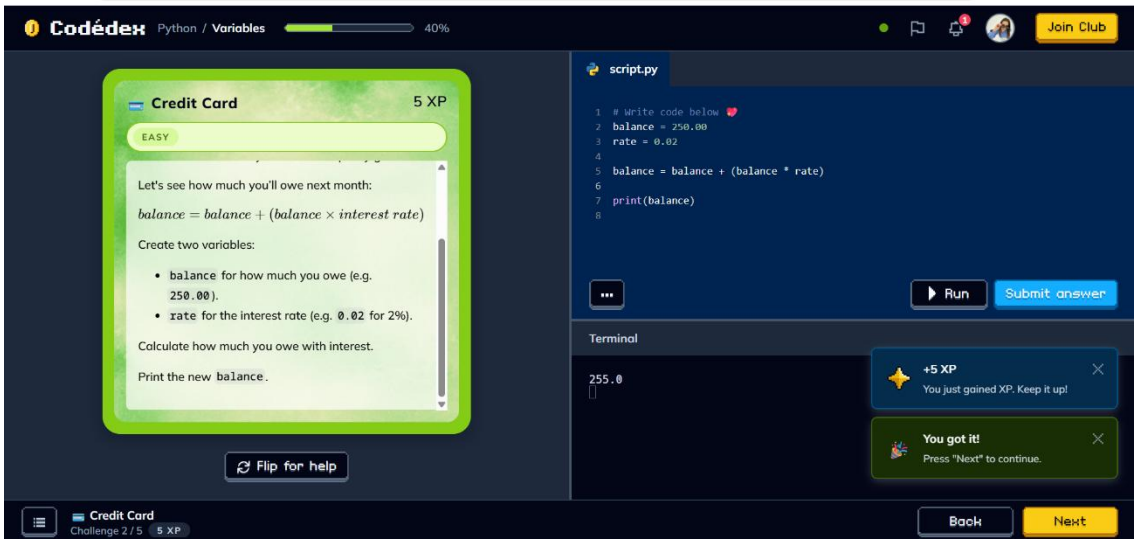
Figura 8 – Am calculat un exercițiu



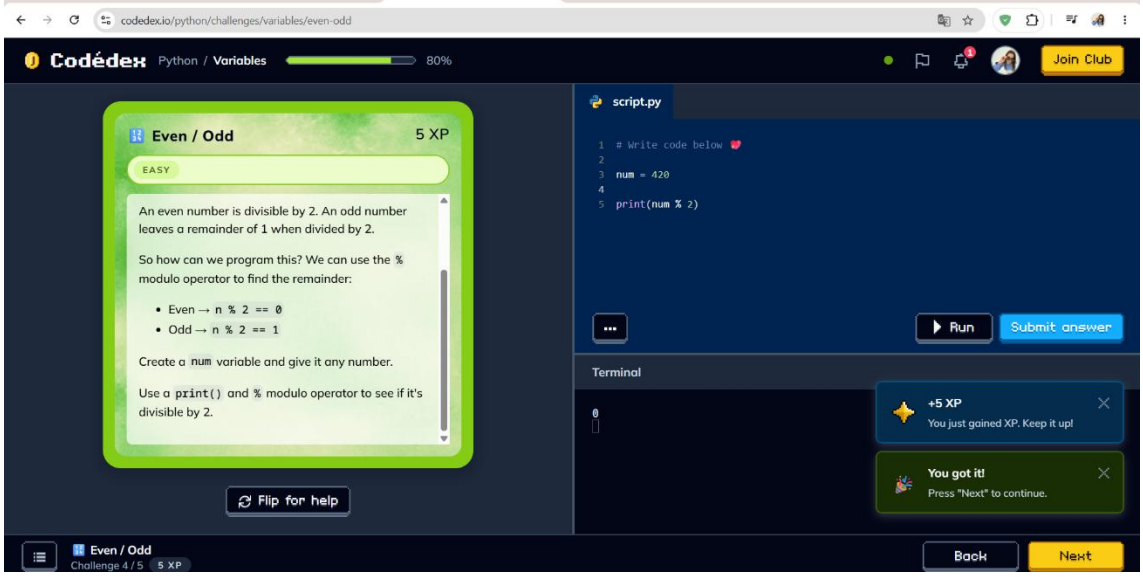
Figura 10 – Calcularea valuta

Erori în Python






Codédex
Python / Variables
60%
Join Club
Moore's Law
5 XP
EASY
Create two variables:
transistors for the number of transistors on your computer right now (e.g. 25000000000).
years for the time into the future (e.g. 10).
To calculate the future number of transistors:
$transistors = transistors \cdot 2^{\frac{years}{2}}$
Print the new transistors.
Flip for help
script.py
Write code below
transistors = 25000000000
years = 10
transistors = transistors * (2 ** (years / 2))
print(transistors)
Run
Submit answer
Terminal
800000000000.0
+5 XP
You just gained XP. Keep it up!
You got it!
Press "Next" to continue.
Moore's Law
Challenge 3 / 5
5 XP
Back
Next


codedex.io/python/challenges/variables/even-odd
Codédex
Python / Variables
80%
Join Club
Even / Odd
5 XP
EASY
An even number is divisible by 2. An odd number leaves a remainder of 1 when divided by 2.
So how can we program this? We can use the % modulo operator to find the remainder:
Even → n % 2 == 0
Odd → n % 2 == 1
Create a num variable and give it any number.
Use a print() and % modulo operator to see if it's divisible by 2.
Flip for help
script.py
Write code below
num = 420
print(num % 2)
Run
Submit answer
Terminal
0
+5 XP
You just gained XP. Keep it up!
You got it!
Press "Next" to continue.
Even / Odd
Challenge 4 / 5
5 XP
Back
Next


Codédex
Python / Variables
100%
Join Club
Pet Age
10 XP
MEDIUM
Then ask the user how old their dog/cat is and store it in an age variable. The user should see:
Enter your dog's age:
Calculate their age in human years and store it in human_age:
1 dog year ≈ 7 human years
1 cat year ≈ 6 human years
Print out human_age.
Flip for help
script.py
Write code below
age = int(input('Enter your dog\'s age: '))
human_age = age * 7
print(human_age)
Run
Submit answer
Terminal
Enter your dog's age: 5
35
+10 XP
You just gained XP. Keep it up!
You got it!
Press "Next" to continue.
Pet Age
Challenge 5 / 5
10 XP
Back
Next

CONCLUZII

Parcurgerea primelor două capitole din Codedex, **Hello World** și **Variables**, mi-a oferit o înțelegere clară a sintaxei de bază și a modul de funcționare a unui program: am învățat să afișez mesaje în terminal cu print(), să lucrez cu tipuri fundamentale de date (int, float, str, bool), să efectuez calcule matematice utilizând operatori aritmetici și de modul (%), și să preiau date de la utilizator prin input(). Toate cele 10 exerciții, articolele și provocările rezolvate mi-au consolidat abilitatea de a structura și manipula informația.

Partea a II -a în Learn Python

Modulul 1: *Learn the Basics*

- Hello,World



Figura 11 – Folosirea print

- Variables and Types

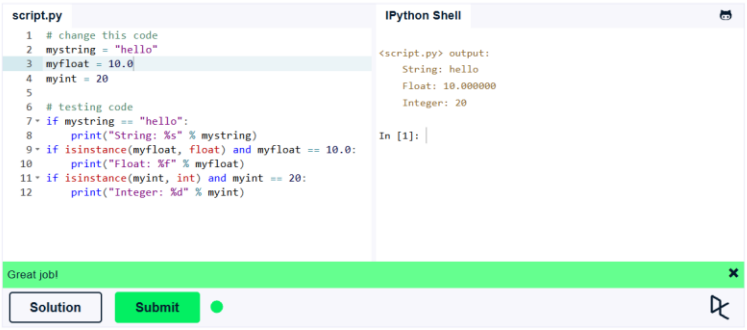


Figura 12 – Utilizarea variabelor

- Lists

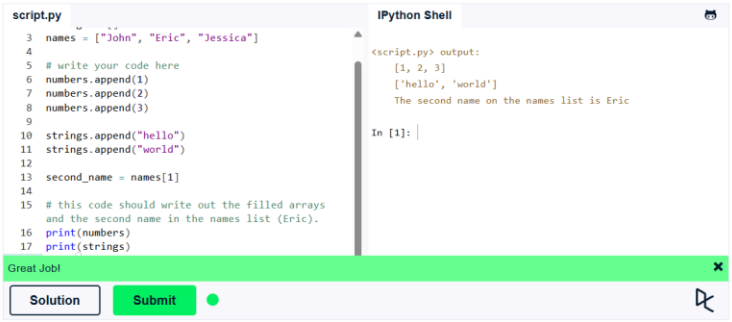


Figura 13 – Utilizarea variabilelor și metoda append

- Basic Operators

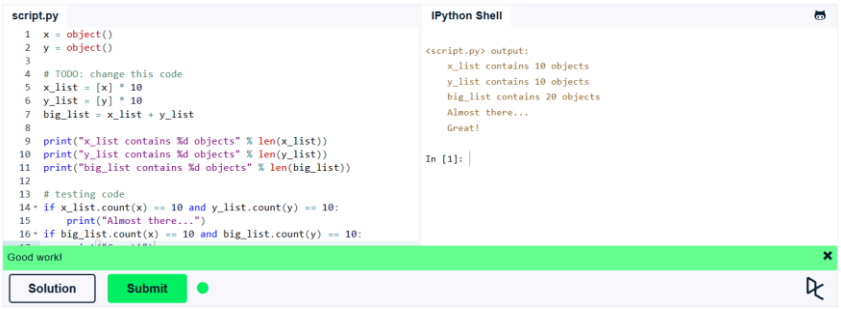


Figura 14 – Utilizarea listei și apelarea ei

- String Formatting

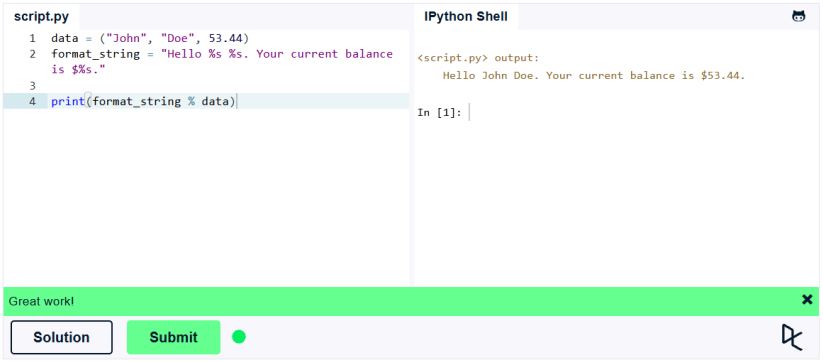


Figura 15 – Utilizarea formatul string

- Basic String Operations

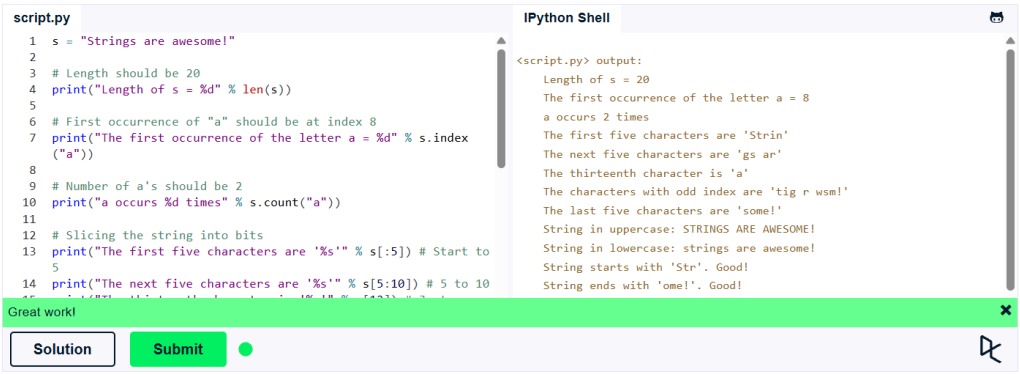


Figura 16

- Conditions

Change the variables in the first section, so that each if statement resolves as True.

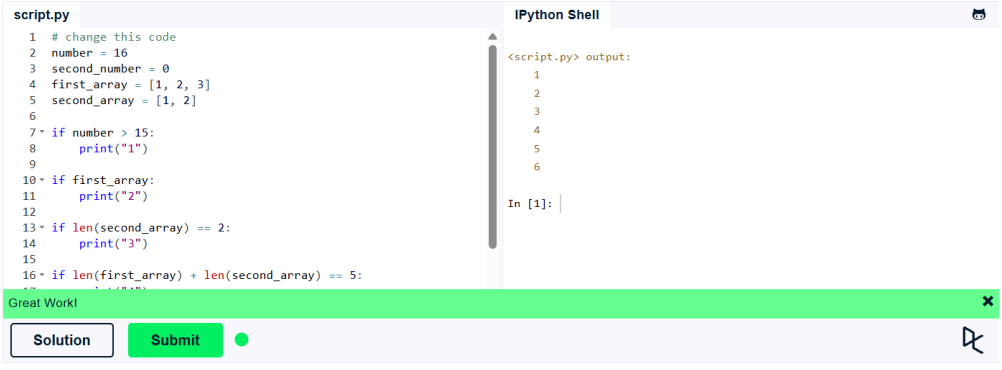


Figura 17

- Functions

Loop through and print out all even numbers from the numbers list in the same order they are received. Don't print any numbers that come after 237 in the sequence.

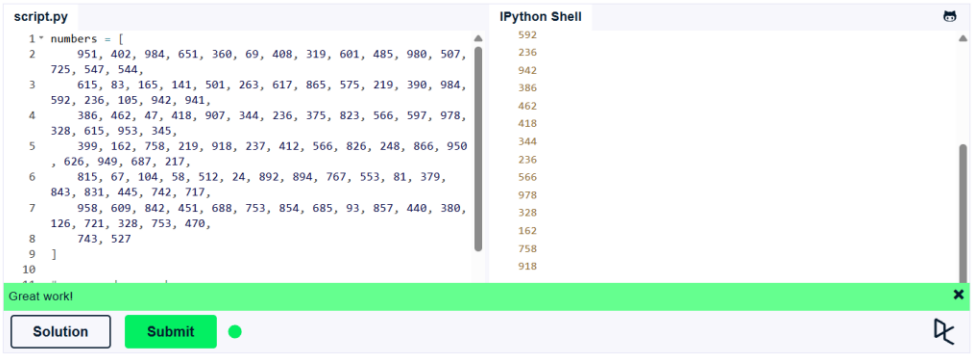


Figura 18

- Classes and Objects

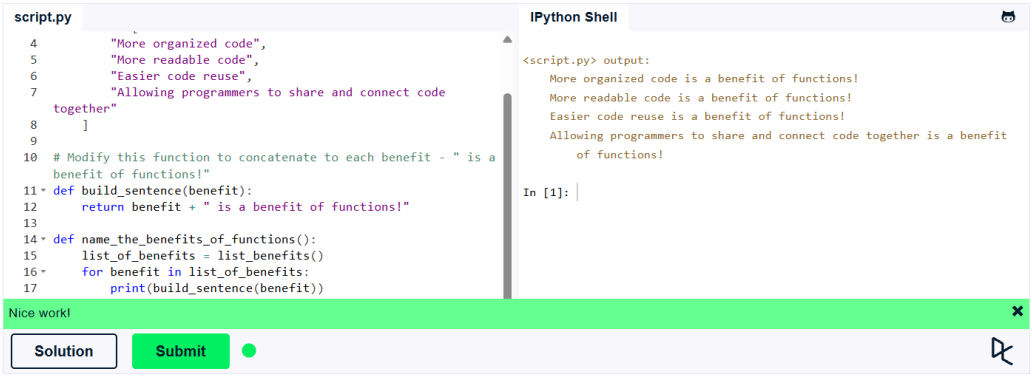


Figura 19

- Dictionaries

We have a class defined for vehicles. Create two new vehicles called car1 and car2. Set car1 to be a red convertible worth $60,000.00 with a name of Fer, and car2 to be a blue van named Jump worth $10,000.00.

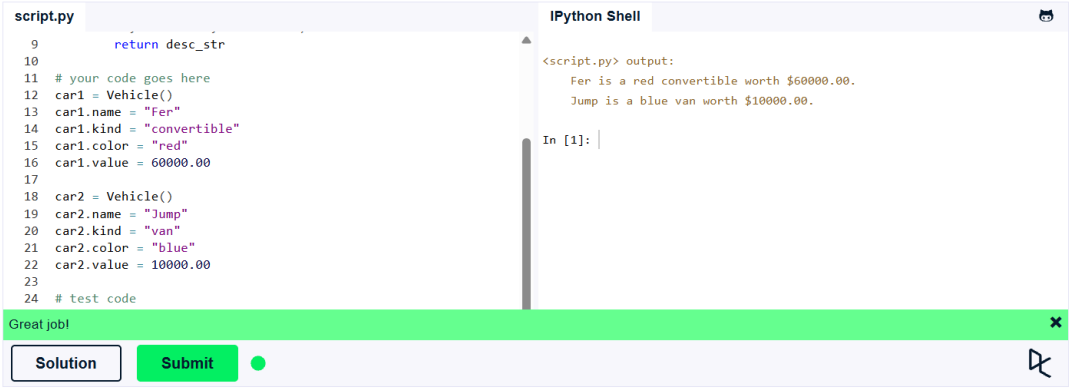


Figura 20

- Modules and Packages

Add "Jake" to the phonebook with the phone number 938273443, and remove Jill from the phonebook.



Figura 21

- Input and Output

In this exercise, print an alphabetically sorted list of all the functions in the `re` module containing the word `find`.



Figura 22

Modulul : *Advanced Tutorials*

Generators

will simultaneously switch the values of a and b.

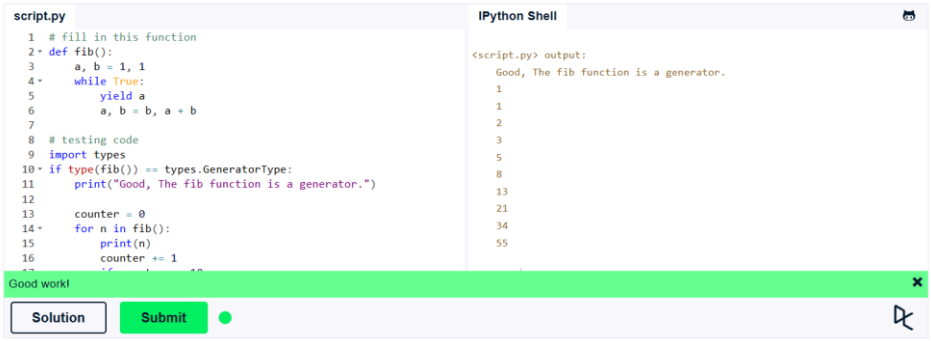


Figura 23

List Comprehensions

Using a list comprehension, create a new list called "newlist" out of the list "numbers", which contains only the positive numbers from the list, as integers.

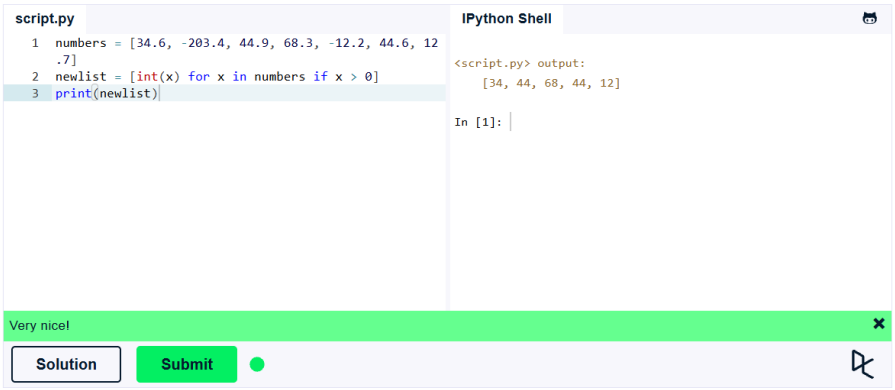


Figura 24

Lambda Functions

Write a program using lambda functions to check if a number in the given list is odd. Print "True" if the number is odd or "False" if not for each element.

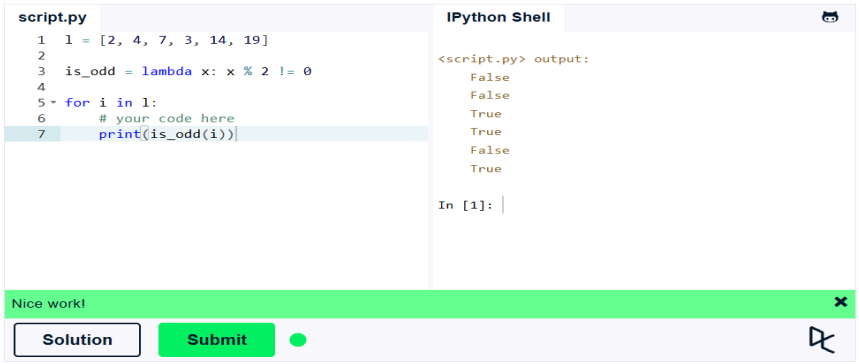


Figura 25

Multiple Function Arguments

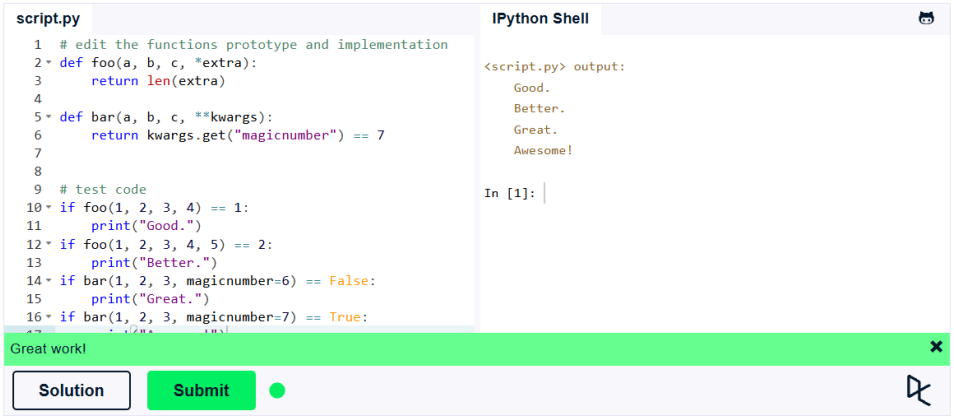


Figura 26

Regular Expressions

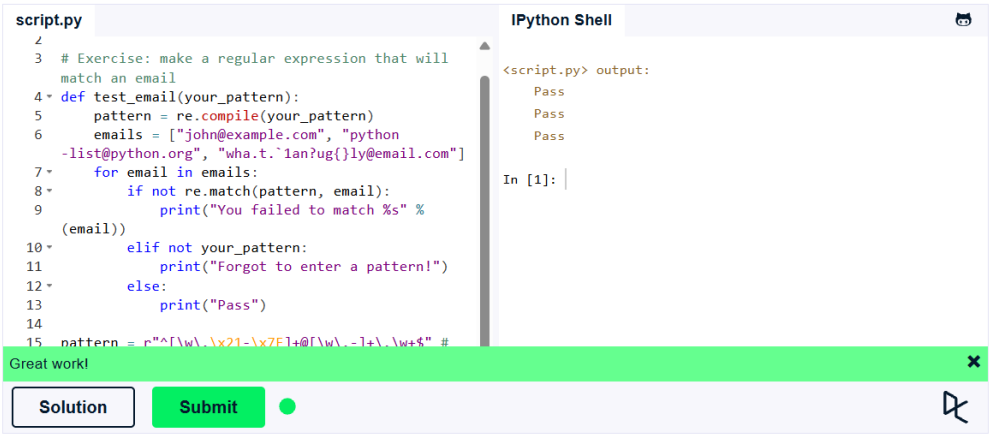


Figura 27

Exception Handling

Handle all the exception! Think back to the previous lessons to return the last name of the actor.



Figura 28

Sets

In the exercise below, use the given lists to print out a set containing all the participants from event A which did not attend event B.

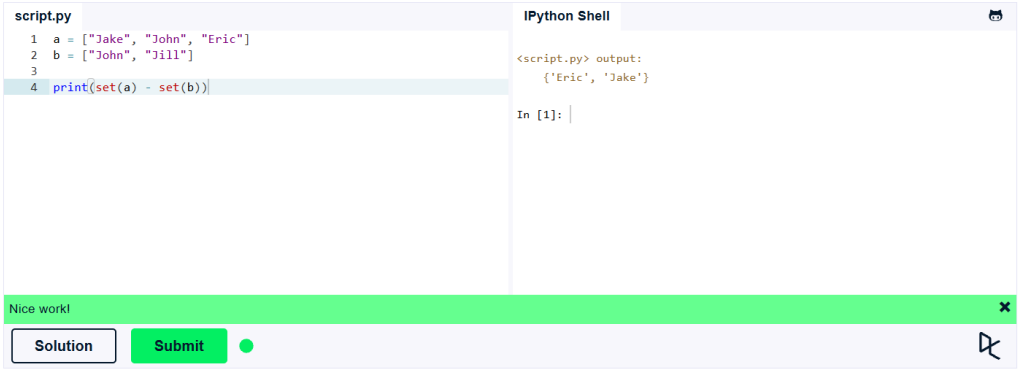


Figura 29

Serialization

The aim of this exercise is to print out the JSON string with key-value pair "Me" : 800 added to it.

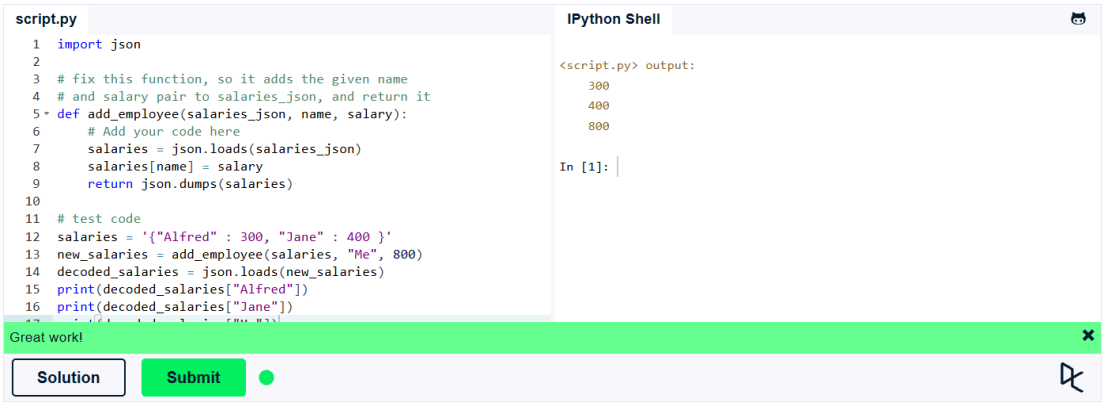


Figura 30

Partial functions

Edit the function provided by calling partial() and replacing the first three variables in func(). Then print with the new partial function using only one input variable so that the output equals 60.



Figura 31

CONCLUZIE

Parcurgerea și rezolvarea cu succes a acestui set de aplicații practice mi-au permis să-mi consolidez cunoștințele de bază și avansate de Python, acoperind subiecte esențiale precum structurile de date, manipularea șirurilor, orientarea pe obiecte și lucrul cu module specifice. Prin îndeplinirea fiecărei sarcini în parte, am demonstrat capacitatea de a scrie un cod curat, eficient și corect structurat, depășind cu succes provocările de sintaxă, gestionare a excepțiilor și logică de programare.

BIBLIOGRAFIE

1. https://www.codedex.io/python
2. https://www.learnpython.org/